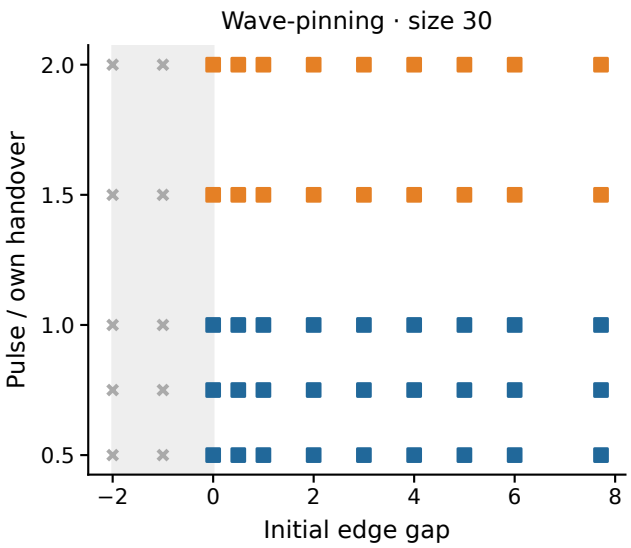
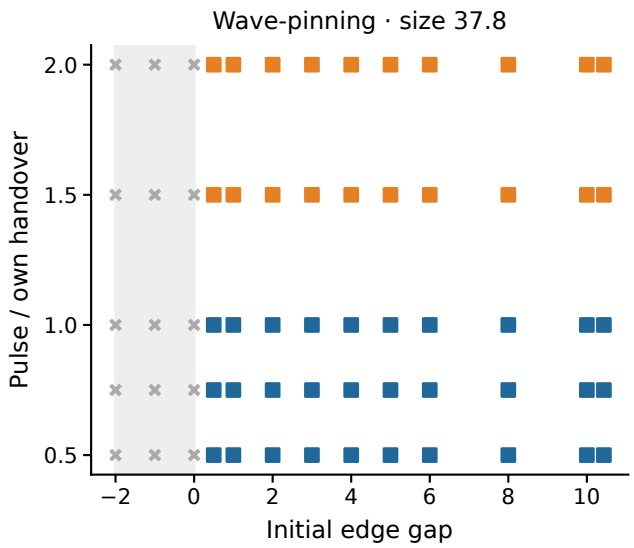
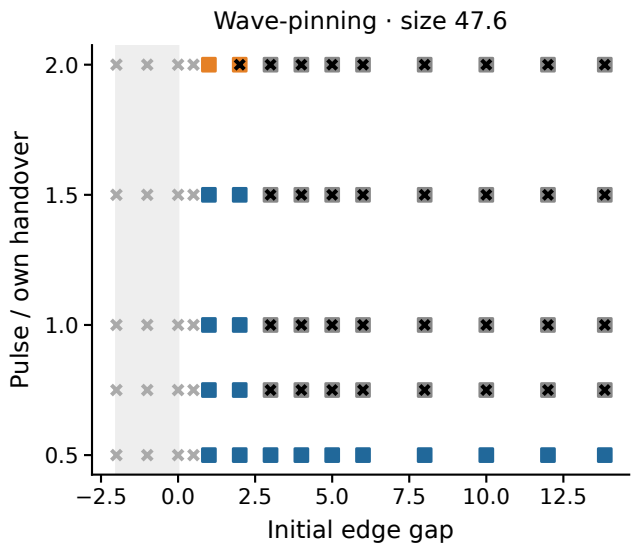
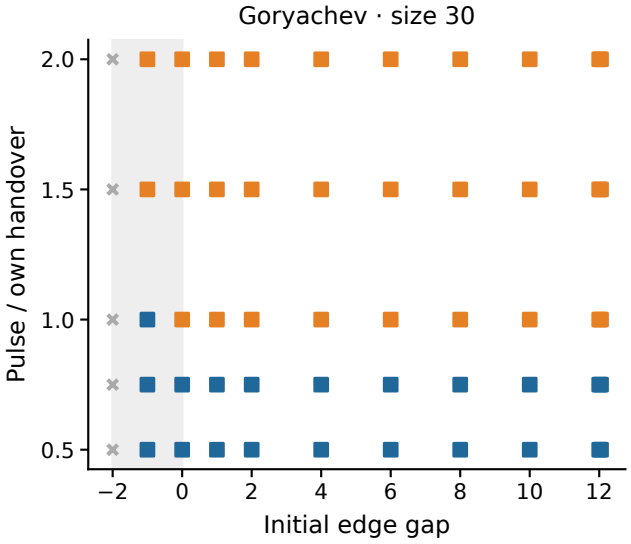
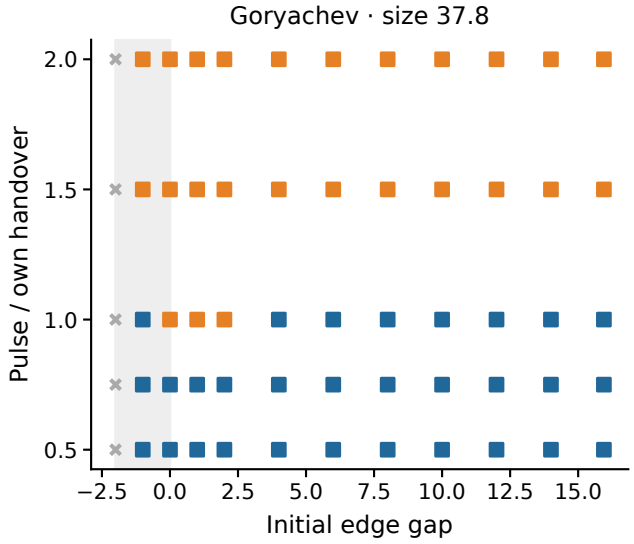
S1 · All 545 cue-withdrawal pulses and 34 unavailable own-handover placements

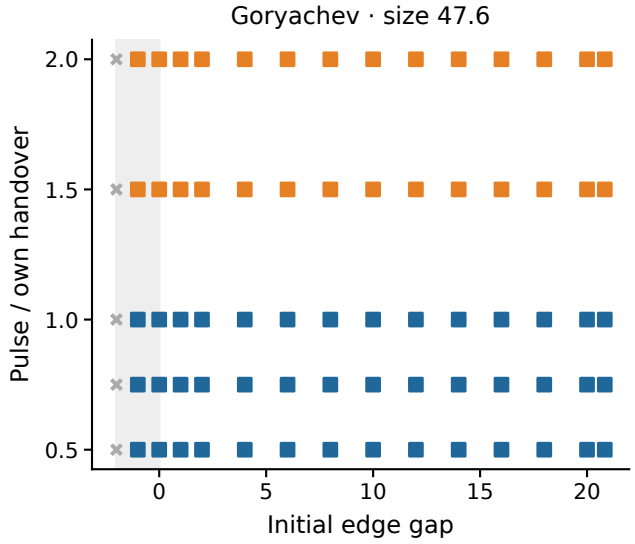


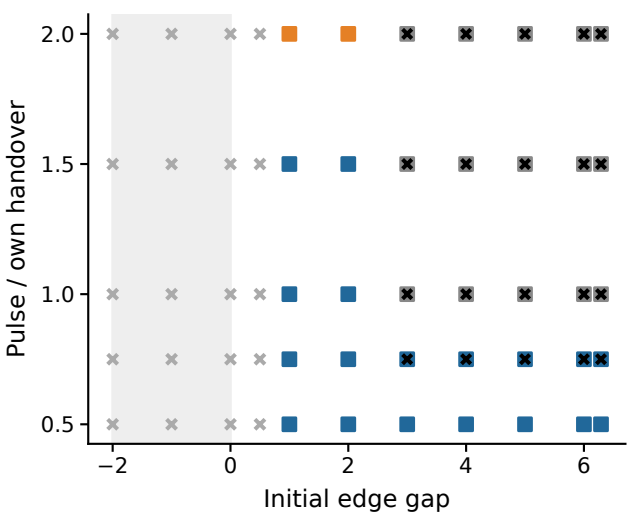


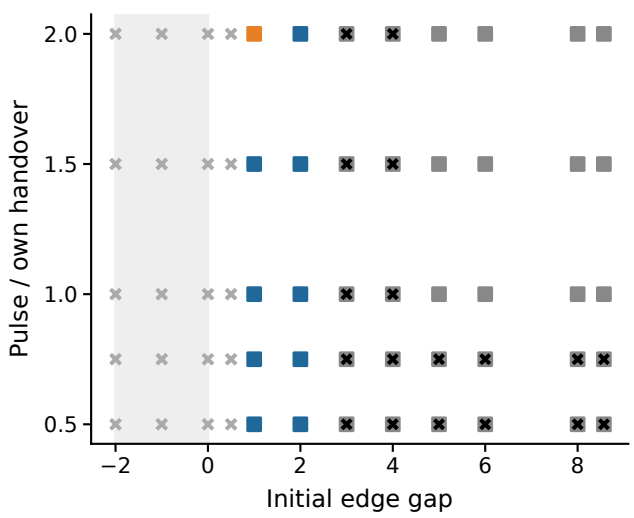


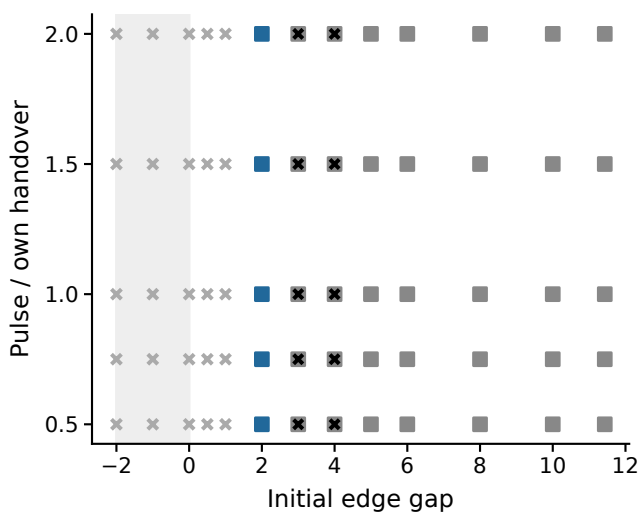


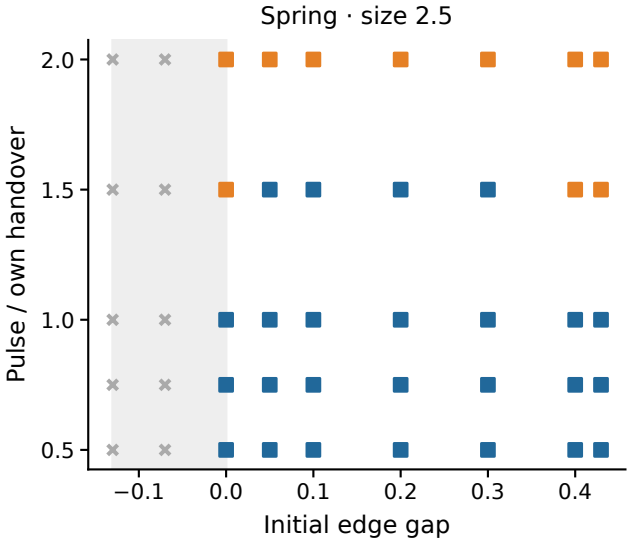


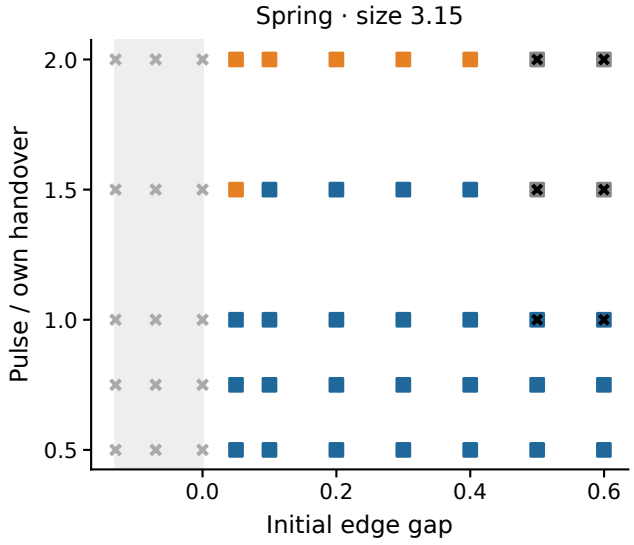


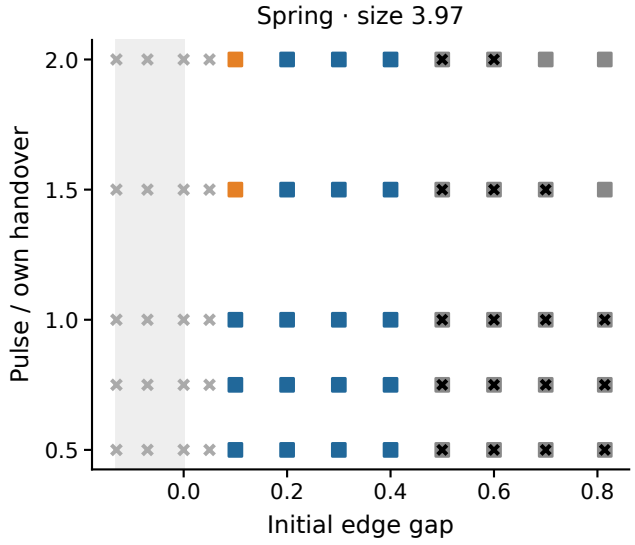








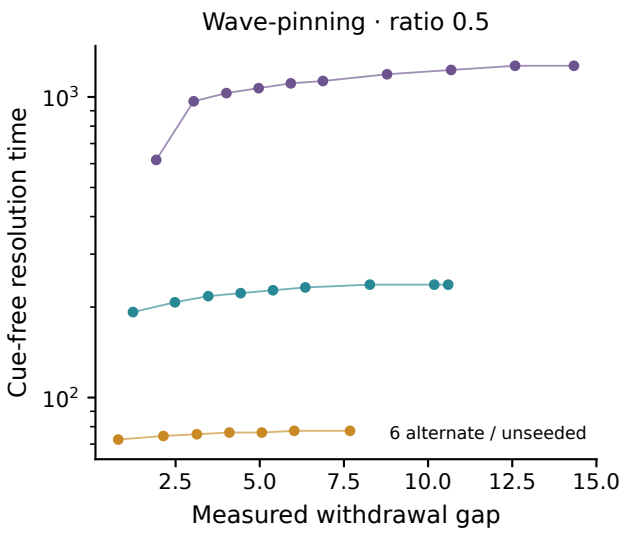


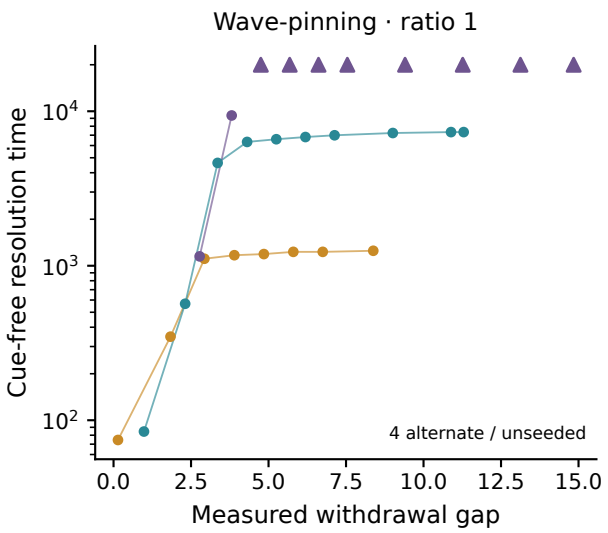


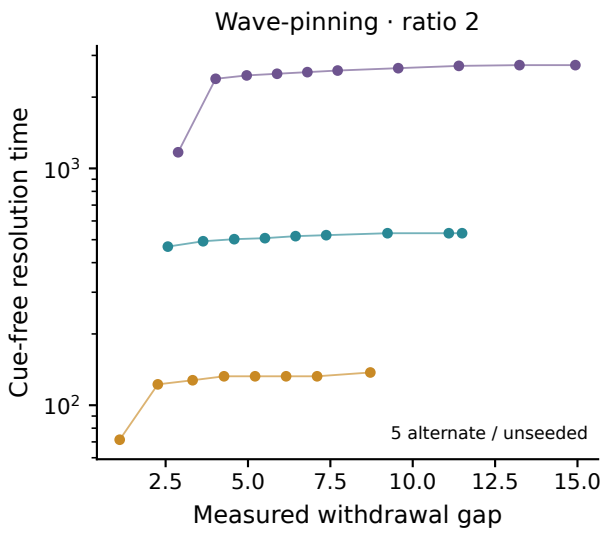
Blue: incumbent-associated final region; orange: challenger-associated; grey: unresolved. Grey ×: no true two-region handover.
Black × over a tested cell: finite native-field tail fails, independent of spatial fate. Shading: borderline placement.

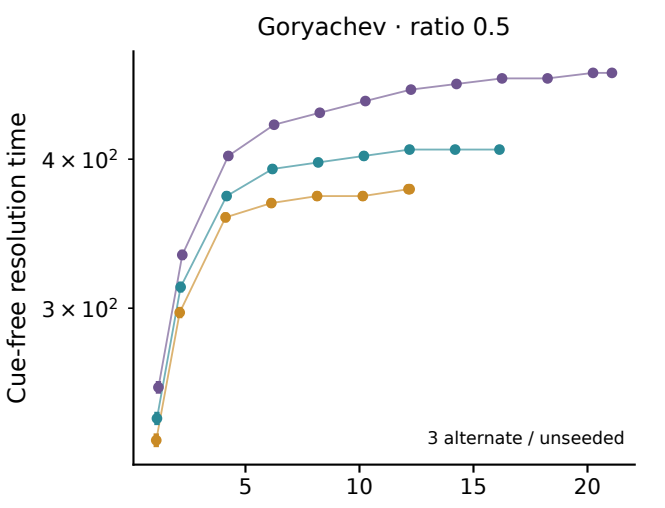
S2 · Cue-free resolution at all three complete-region withdrawal ratios

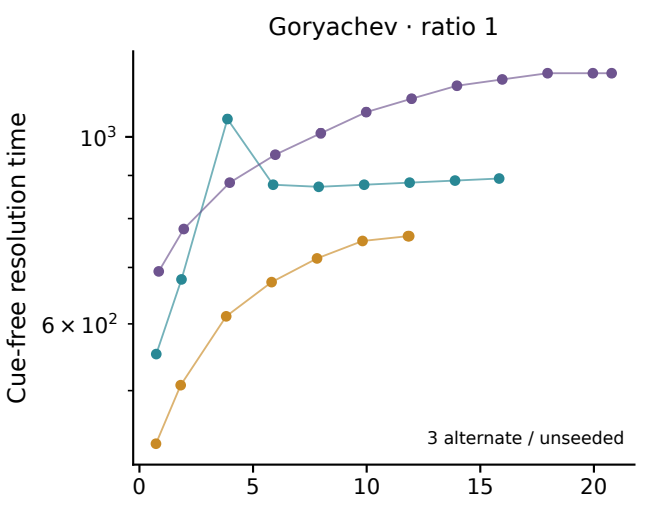
reference size · 1.26× native size label · 1.59× native size label

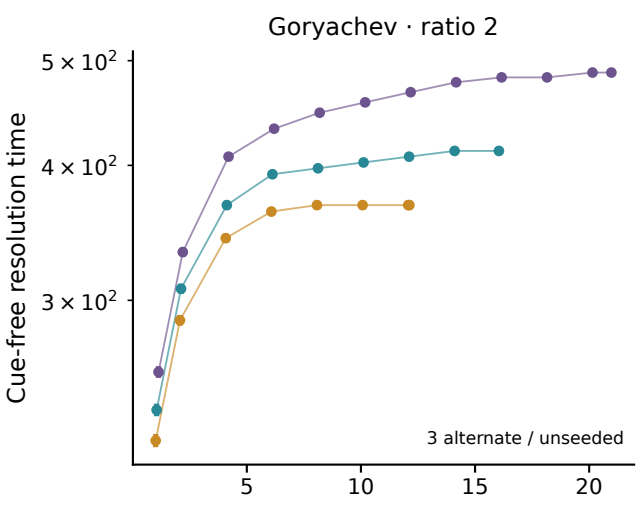


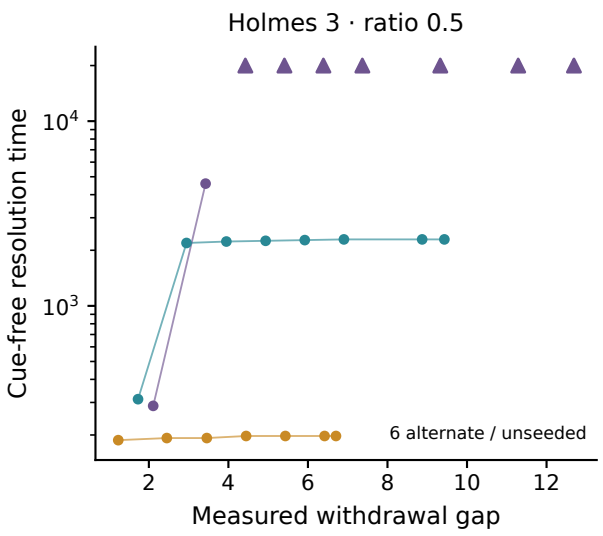


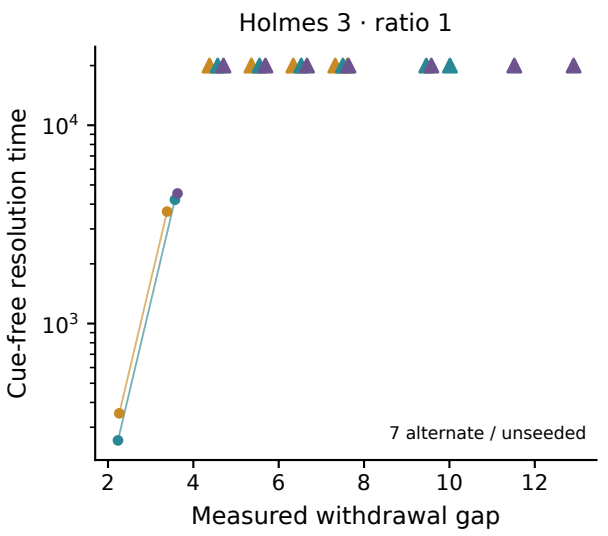


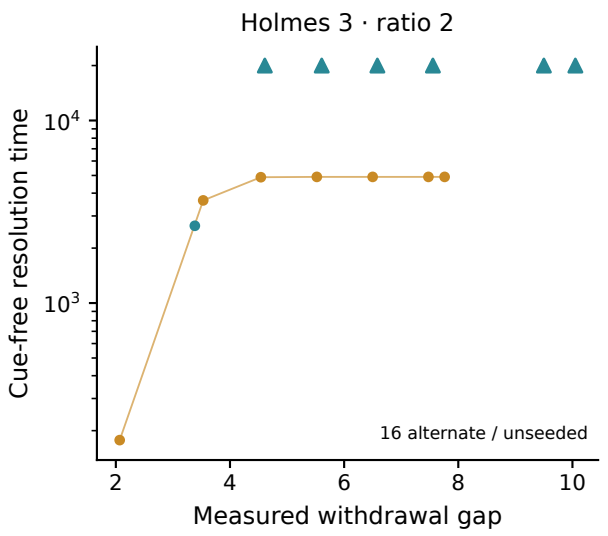


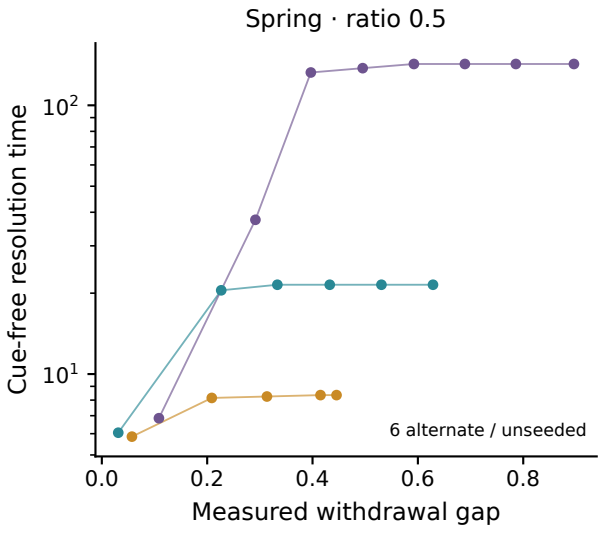


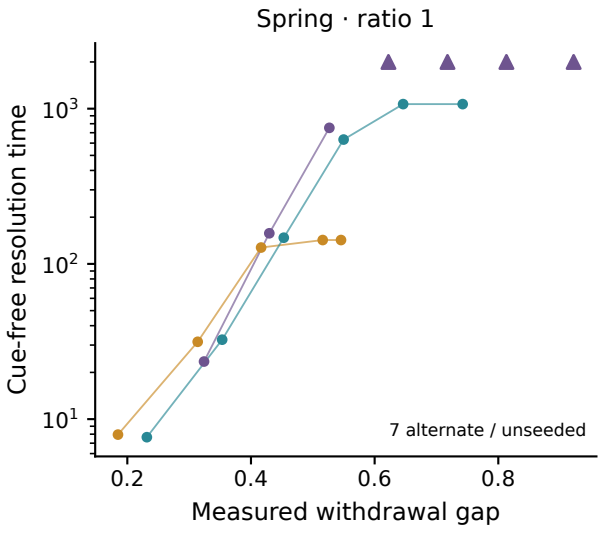


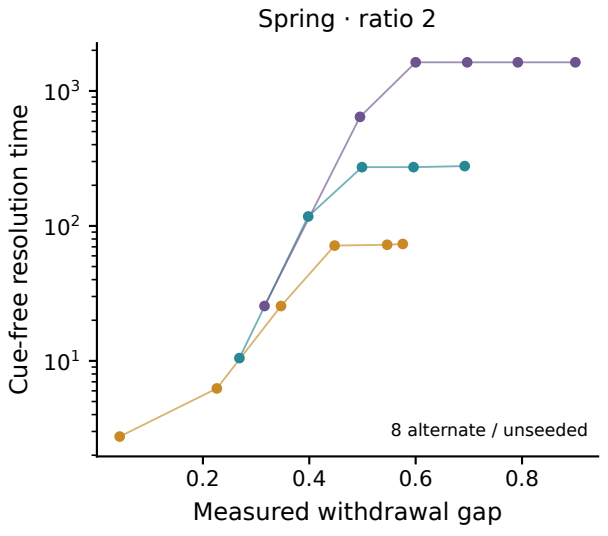




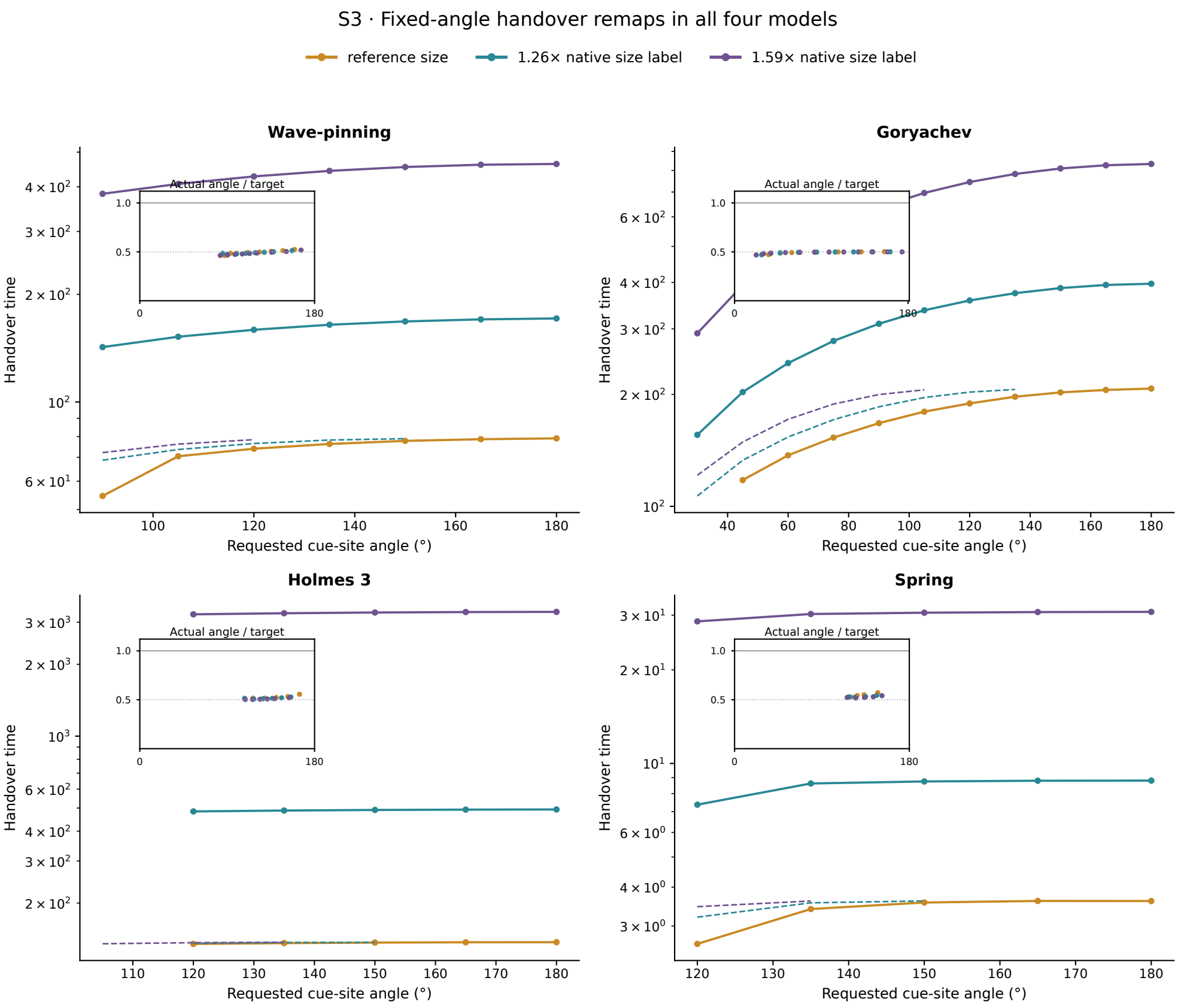








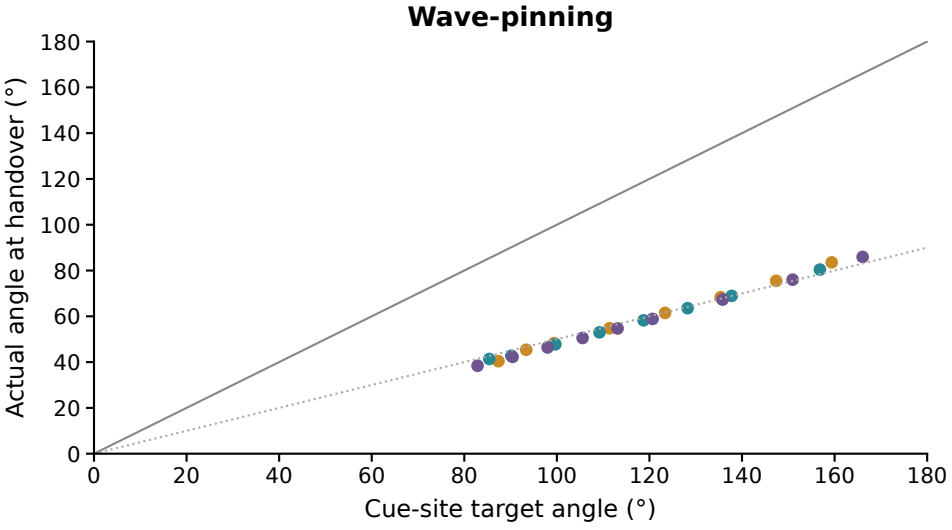
Primary gap: current global10%-contrast contours. ▲: resolution-time lower bound at completed horizon.
Alternate, merged and unseeded cases remain in the complete evidence table and native-history reader.

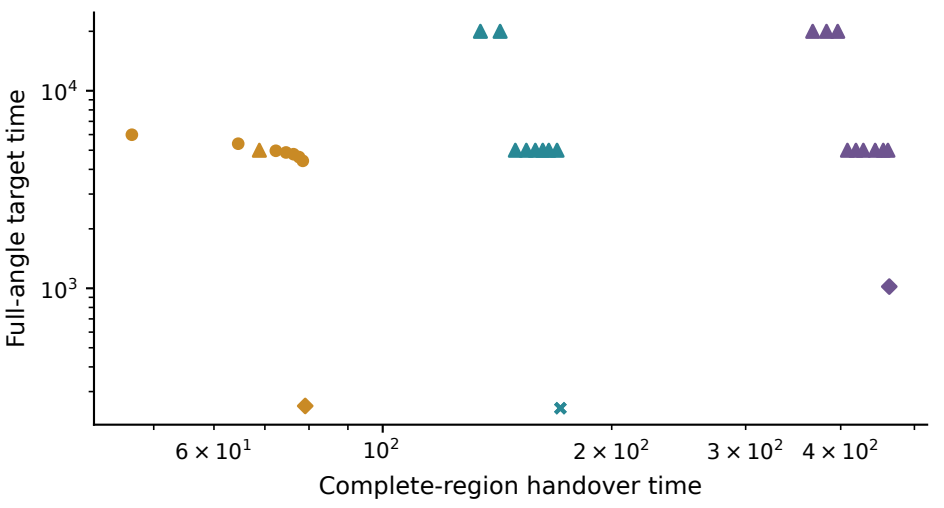

S3 · Fixed-angle handover remaps in all four models
reference size
1.26× native size label
1.59× native size label
Wave-pinning
Goryachev
Holmes 3
Spring
Handover time
Requested cue-site angle (°)
Actual angle / target
Solid: measured native size; dashed: reference-size handover curve evaluated at the larger cell's measured-width gap.
No extrapolation or interpolation through unavailable/censored neighbors. Insets: actual direction at handover / requested cue angle.

S4 · Handover typically reaches part of the requested angle; full-target timing is separate

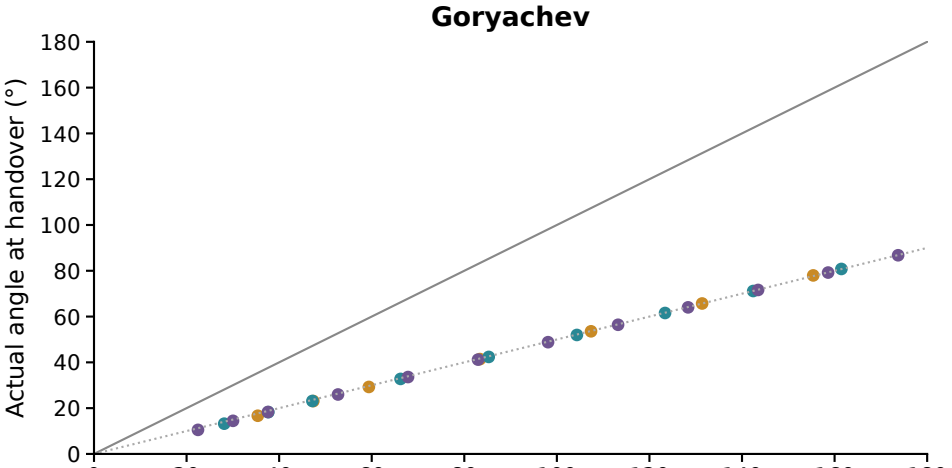
reference size · 1.26× native size label · 1.59× native size label

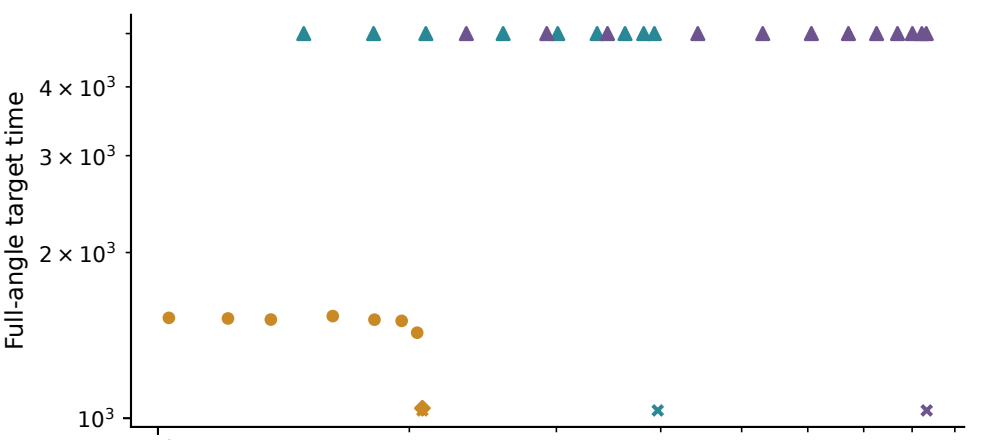
Wave-pinning



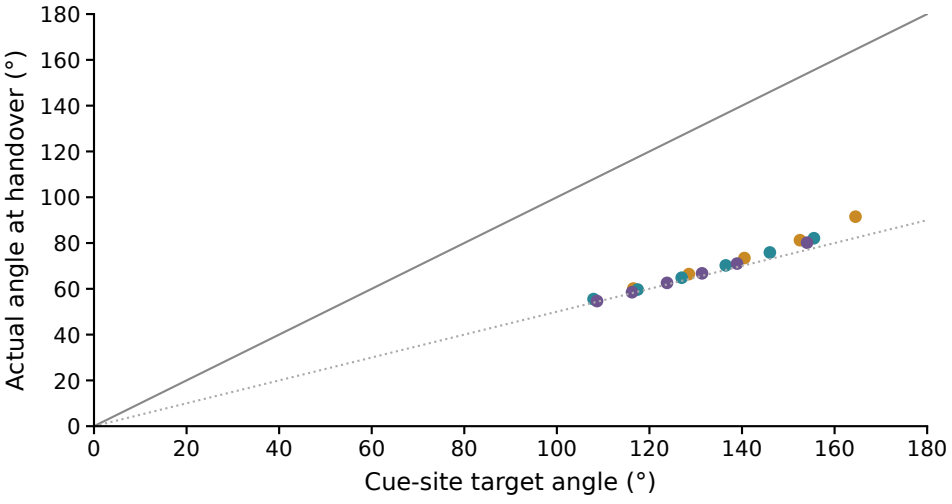


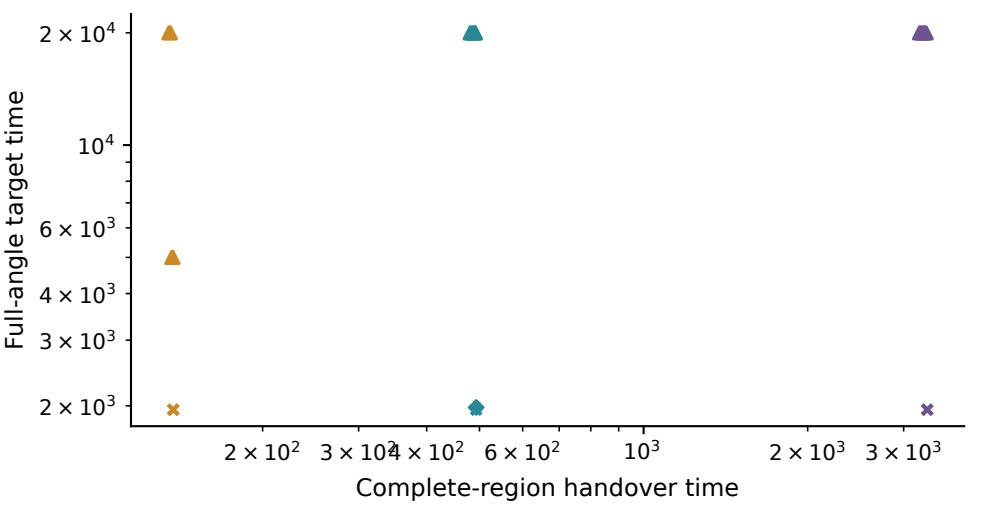
Goryachev



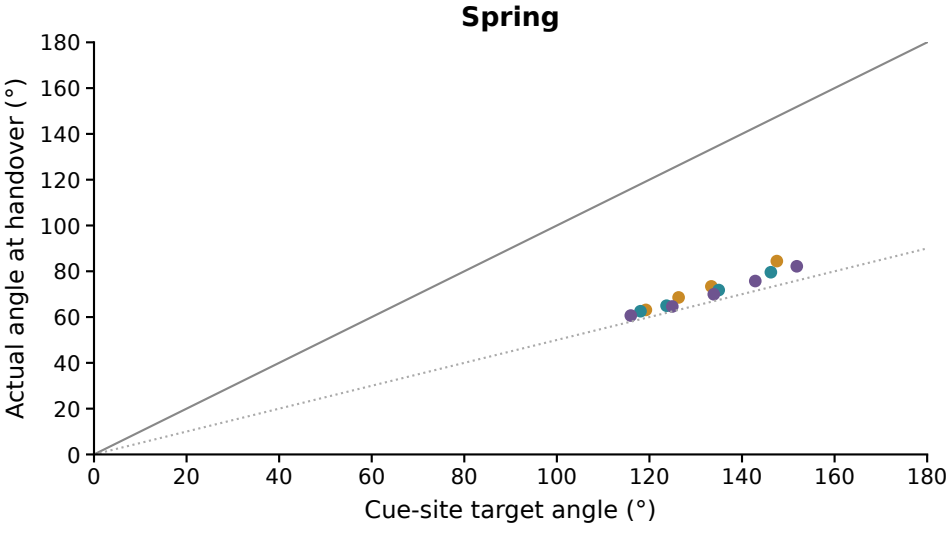


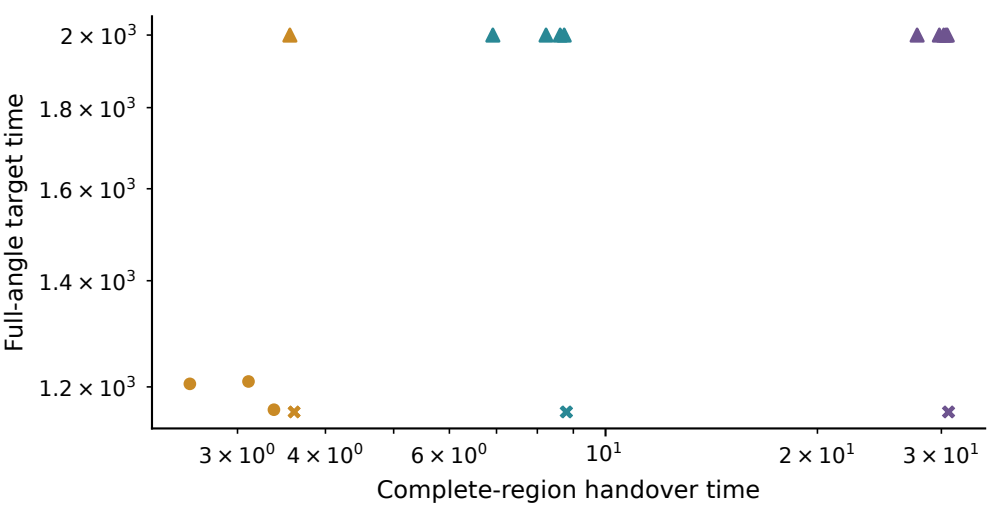
Holmes 3



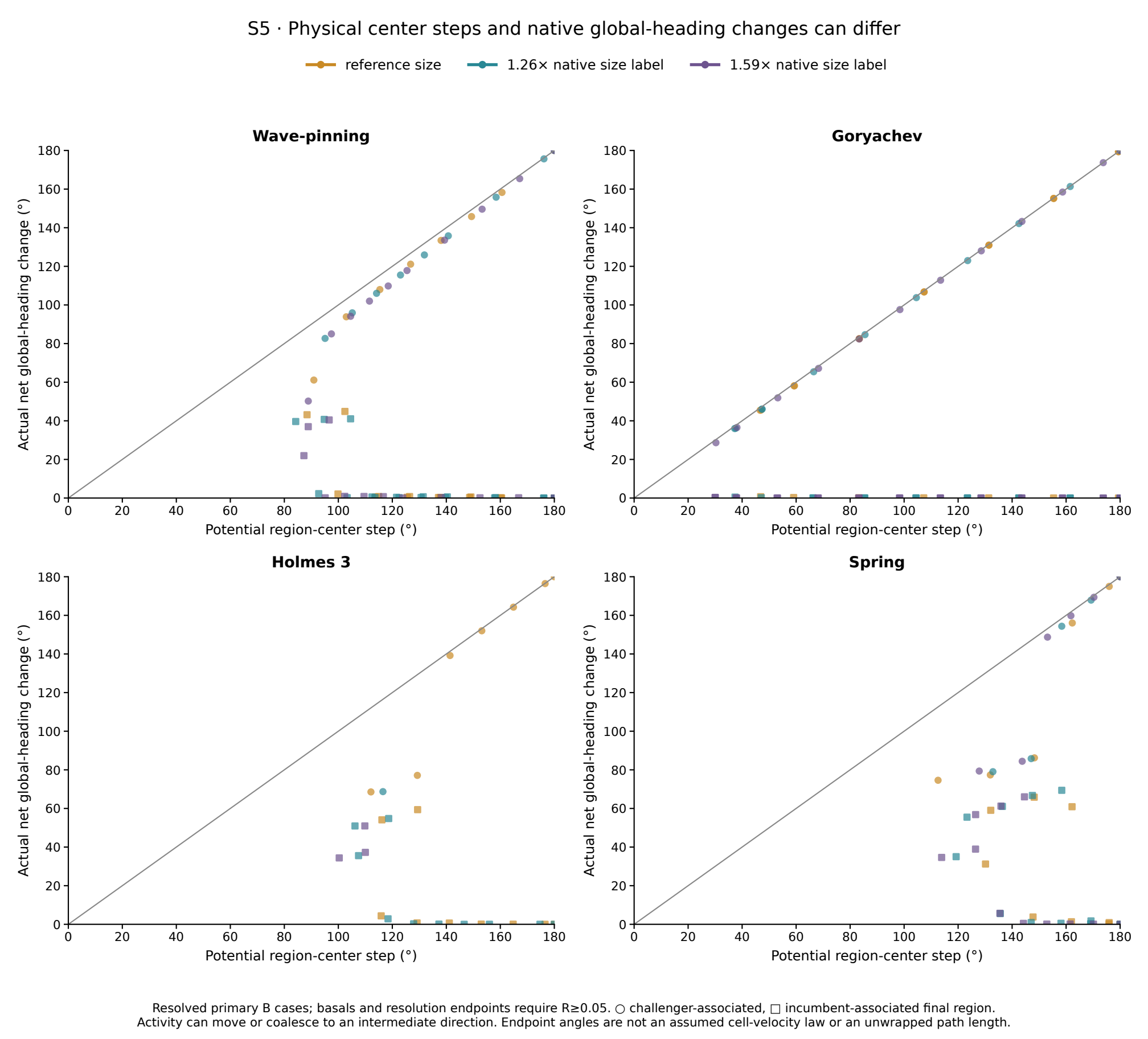


Spring

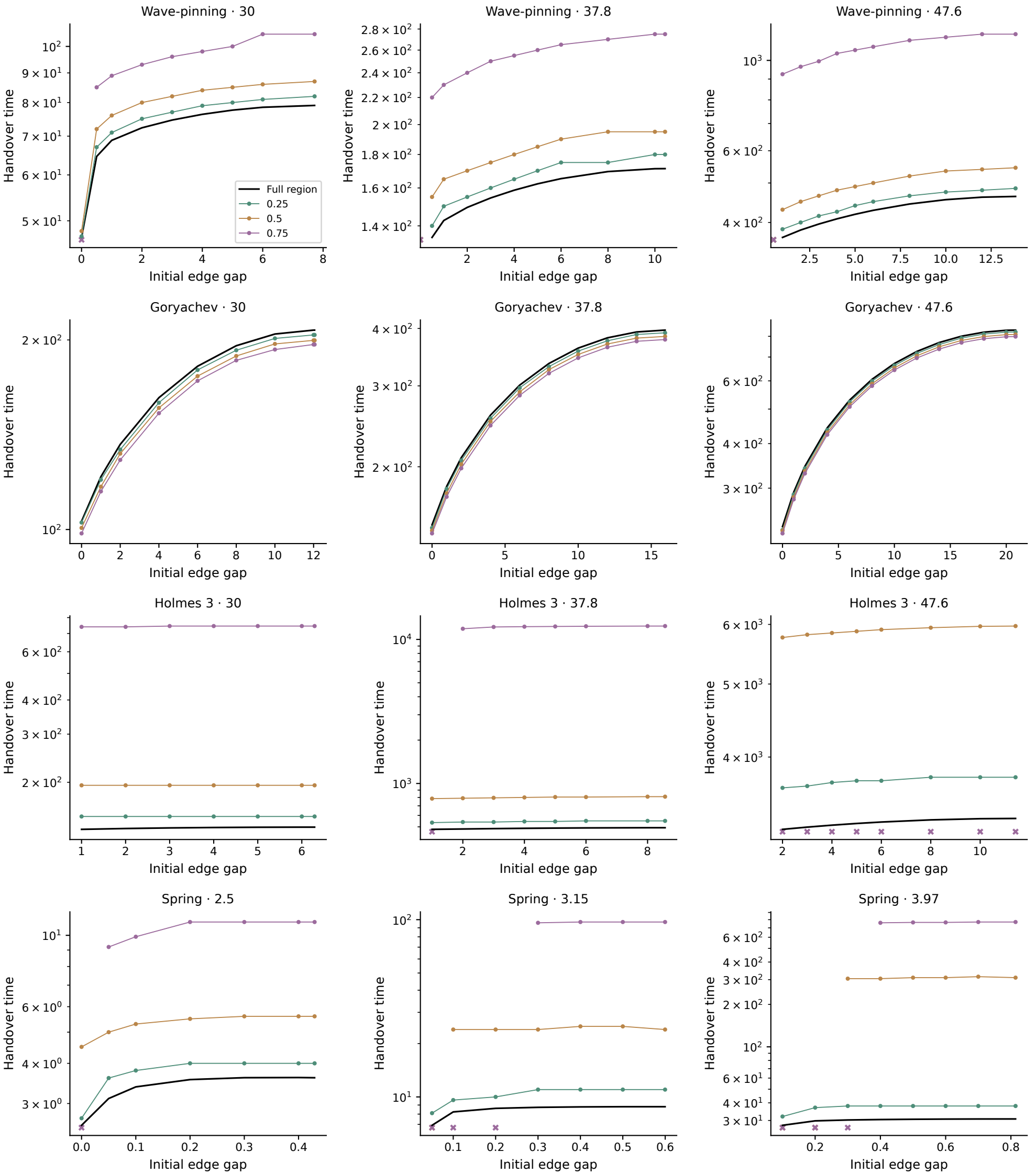




R=|M₁|/M₀≥0.05 at usable samples and intervening conservative complex-moment chords. ▲: finite-cap timing lower bound.
×: direction-quality/cancellation gap prevents a full timing bound. Exact-direction and0.1/0.2 gates plus50/90/99% targets are tabulated.

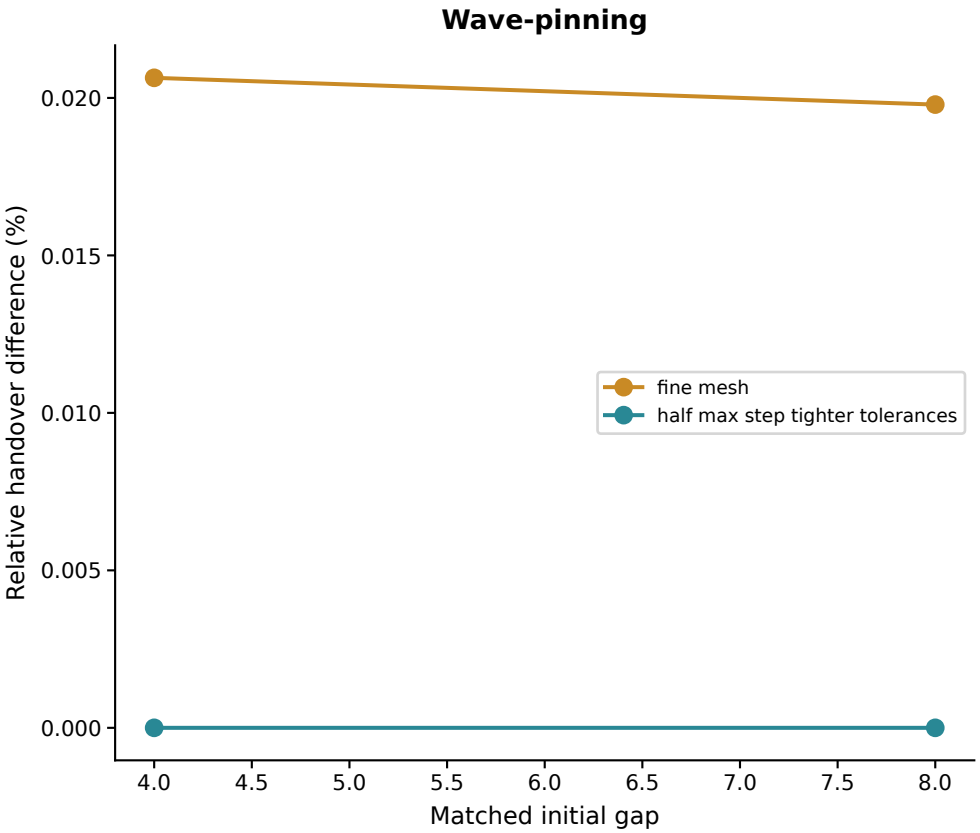

S5 · Physical center steps and native global-heading changes can differ
reference size
1.26× native size label
1.59× native size label
Wave-pinning
Goryachev
Holmes 3
Spring
Actual net global-heading change (°)
Potential region-center step (°)
0
20
40
60
80
100
120
140
160
180
Resolved primary B cases; basals and resolution endpoints require R≥0.05. ○ challenger-associated, □ incumbent-associated final region.
Activity can move or coalesce to an intermediate direction. Endpoint angles are not an assumed cell-velocity law or an unwrapped path length.

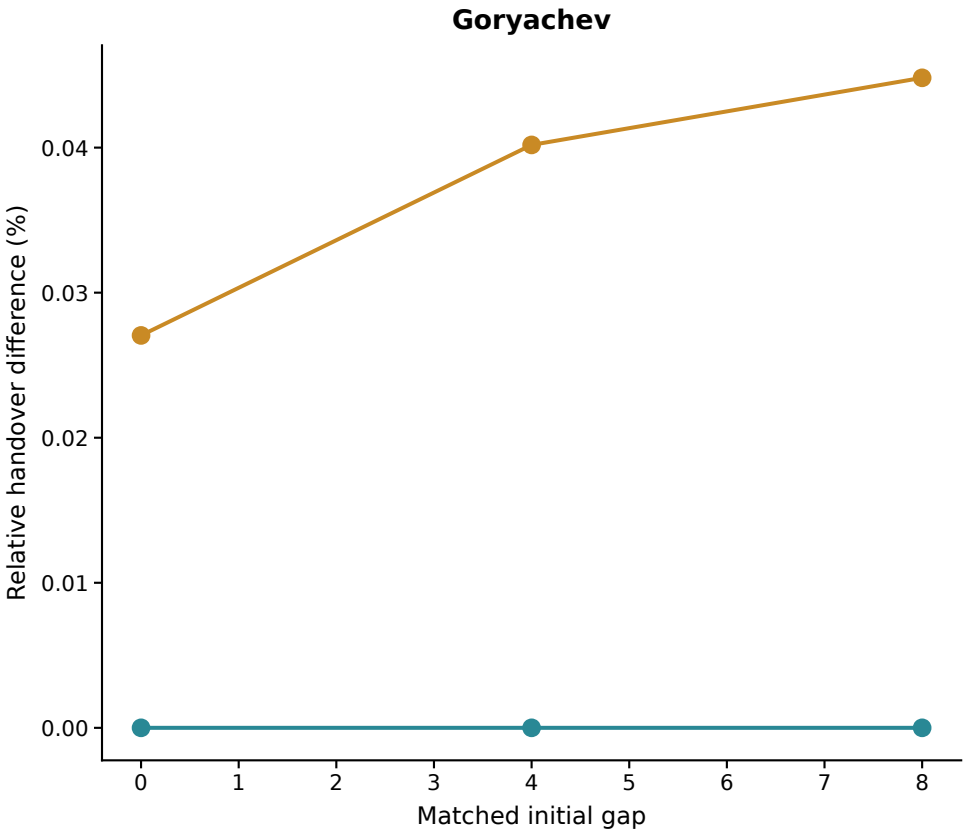
S6 · Amount sensitivity: local 25/50/75% peak contours versus complete-region excess



Original excess above each region's local-peak contour; watershed basins and common current floor are unchanged.
Times are retained native-sample upper brackets. ×: unavailable continuous sampled two-region crossing; no cap substitution.

S7 · Numerical, preparation and complete-source checks





Numerical evidence

Fresh native basals: 12 / 12 qualified
Long no-cue controls: 53,000 chemical / 5,300 Spring
Fine-mesh basals: 2 / 2 independently qualified
A1 refinement controls: 12 / 12 native audits
Observed control roots: 10 / 10 verified
A2 native audit: 545 / 545
B native audit: 381 / 381 (24 mirrors agree)
B mesh/step controls: 36 / 36 native audits

Readout and source gates

Native positivity and conserved-pool gates pass.
Scalar fate and field stationarity remain separate.
Continuous roots replay the original upper fields.
B controls retain all seed statuses and winners.
35 / 36 resolution brackets overlap.
Goryachev g=4, ratio 1, fine mesh: 1040–1060
versus 1080–1100 (about 4% timing sensitivity).
Full native atlas: 1,150 histories; all sources linked.